Mahima Srivastava

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Education

Chemical Engineering, Ph.D.
University of Maryland College Park — Aug 2021 – May 2026 (Expected)
GPA: 4.0/4.0
Chemical Engineering, B. Tech.
Indian Institute of Technology Kanpur — Jul 2015 – Jun 2019

Professional Summary

Polymer and formulation scientist with 4+ years managing projects in drug delivery and advanced materials. Experience with chemical synthesis and stability studies, characterization, device-drug prototypes, and modelling and data analysis, collaboration, design of experiments, grant development and SOPs, and commercialization pathways.

Skills

- **Materials:** Polymers, hydrogels, biologics, vesicles, liposomes and lipid nanoparticles, surfactants, cell culture
- **Characterization:** Rheology, UV-Vis, DLS/Zetasizer, SEC, HPLC, FTIR, SEM-EDS, MS, Fluorescence Spectroscopy, TGA/DSC, Flow Cytometry, Ultracentrifuge, SDS-PAGE
- **Technical:** Python, Java, C++, COMSOL, MATLAB, ANSYS, ImageJ, Origin, Autodesk, Microsoft Office

U.S. Patent and Invention Disclosure

- Plastic products that can be rapidly degraded on-command. U.S. Patent App. No. 63/679,017 (2025)
- Soft materials that absorb impact. U.S. Patent App. No. 18/640,988 (2024)

Research Experience

Doctoral Research Candidate, University of Maryland — Jan 2022 – Present

- **Zero-Order Drug Delivery:** Engineered skin-covered hydrogel beads achieving 7-day zero-order release of several commercial drugs, validated using UV-vis, HPLC, and Franz cell diffusion; tuned skin chemistry to modulate kinetics and diffusion.
- **Imaging Biomarker for PDAC:** Designed avidin-biotin-functionalized liposomal MRI contrast agents carrying biotinylated anti-CA-19-9 antibodies and gadolinium; used SEC to separate antibody-conjugated liposomes from free antibodies.
- **Ultra-Shear-Thinning Hemostatic Gel:** Designed new hemostatic gels with fine-tuned rheological properties, ensuring that the gel is strong and robust (G' >100 Pa) and remains injectable through long(>2m), narrow (<2mm diameter) catheters.
- **On-Demand Degradable Plastics**: Synthesized novel on-demand degradable bioplastics with excellent tensile and torsional properties, achieving complete degradation in 2 minutes at room temperature; currently optimizing crystallization process for scale-up from bench-scale to pilot production.
- **Tissue Mimic:** Performed comprehensive rheology and mechanical characterization of animal tissues in small and large deformation regimes; developed hydrogels with tunable viscoelastic properties to mimic native tissues.
- **Impact-Absorbing Materials:** Developed eco-friendly and sustainable gels capable of absorbing impact forces for protective applications in fragile material packaging; designed and validated scale-up protocols, optimizing mixing sequences, gelation kinetics, and process parameters to achieve batch-to-batch consistency at pilot scale.
- **Stimuli-Triggered Organogel Degradation:** Developed DBS organogels that can be degraded with stimuli, with programmable lifetimes (minutes-days); used NMR and MS to explain acid-catalyzed degradation mechanisms.
- **Intranasal Drug Delivery**: Formulated CO_2-triggered surfactant gel-to-sol system integrated into a wearable device for intranasal naloxone; demonstrated reversibility and drug stability across 10 cycles using FTIR; characterized different stages using cryo-TEM and DLS; collaborated with a medical technology startup to develop prototype, prepared submission strategy for FDA Type C consultations.
- **Fast-expanding Porous, Degradable Hydrogels:** Modified chitosan by attaching hydrophobic groups on the chain and used them to produce porous hydrogels with 100x expansion in < 30 s and on-demand degradation in 30 min using ambient triggers; quantified pore-size distribution using SEM and swelling ratio by gravimetry.

Undergraduate Research Assistant, IIT Kanpur — Jul 2018 – Jun 2019

- **Granular Flows Modeling**: Performed simulations for viscoplastic fluid models in COMSOL and discrete element method (DEM) simulations for 2D, steady-state, granular flow dynamics on inclined surfaces, achieving strong correlation between continuum and particle-based models.

Undergraduate Research Assistant, IIT Bombay Dec 2016 – Jan 2017

- **Identifying Binding Pockets of Sialic-Acid Binding Proteins**: Designed the algorithm to read protein data from RCSB database and identify the probable binding sites of sugar, in Python; verified the classification using SCOP and CATH.

Work Experience

- **Programming Analyst,** Citicorp Services Pvt. India. Ltd., Pune, India Aug 2019 – Aug 2021
  Built and deployed Docker-containerized Java and Python applications on Kubernetes, creating a scalable and reproducible pipeline for automated data analysis.

Publications

- **Srivastava, M.**, Choudhary, H., Varisco, D., Hsu, M., Chen, P.Y., Goodman, R. N., & Raghavan, S.R. Plastic Products that Can be Rapidly Degraded On-Demand. *Submitted*
- Colton, A., Halli, R.N., Ma, M.R., Nori, T., Muller, L.K., Barvenik, K.J., **Srivastava, M**., Ramdam, B., Sarker, S., Tubaldi, E. and Kofinas, P., 2025. Geometric determinants of sinterless, low-temperature-processed 3D-nanoprinted glass. Microsystems & Nanoengineering, 11(1), p.145.
- Borden L. K.; Nader, M.; Burni, F.; Grasso, S.; Ortega, I.; **Srivastava, M**.; Atienza, P. M.; Erdi, M.; Wright, S.; Sarkar, R.; Sandler, A.; Raghavan S.R; Switchable Adhesion of Hydrogels to Plant and Animal Tissues. Advanced Science, 2411942.

Leadership & Teaching

Instrumentation Training & Support Jan 2022 – Present

- Trained new lab members on core laboratory instruments (Rheometer, SEM, DLS, UV-vis, Tensiometer, and Instron).
- Developed SOPs to ensure consistent operation, troubleshooting and data quality.

Technical Advisor – Patent Commercialization May 2025 – Present

- Collaborated with business development teams to assess market opportunities, scale-up requirements, and technical feasibility for impact-absorbing materials and degradable plastics.
- Provided technical guidance and trained teams on materials' synthesis and scale-up protocols.

Laboratory Safety Officer Jan 2022 – Present

- Authored and enforced chemical hygiene/SOPs for lab equipment and conducted hazard assessments.
- Trained lab members on waste management and PPE for 4+ years and maintained compliance with institutional EHS.

Research Mentor Jan 2022 – Present

- Supervised five undergraduates in our lab group and helped 3 of them to secure financial aid through UMD ASPIRE program.
- Received formal training on being a mentor and mentored a high school student as a part of Center for the Improvement of Mentored Experience in Research (CIMER).

Graduate Teaching Assistant Aug 2022 – May 2023

- Provided instructional support and led discussions and tutoring sessions for Bionanotechnology and Process Dynamics and Controls.

Conference Presentations

- **Srivastava, M.**, Nader, M., Varisco, D., & Raghavan, S.R.
  Can hydrogels Mimic the Rheology of Animal Tissues?
  Pacifichem 2025, Honolulu, HI
- **Srivastava, M.**, Choudhary, H., Varisco, D., Hsu, M., & Raghavan, S.R.
  Bioplastics that Can be Degraded On-Command within Minutes.
  Pacifichem 2025, Honolulu, HI
- **Srivastava, M.**, & Raghavan, S.R.

Designing Gels That Mimic the Rheology of Animal Tissues.
2025 AIChE Annual Meeting, Boston, MA

- **Srivastava, M.**, Choudhary, H., Varisco, D., Hsu, M., & Raghavan, S.R.
  On-Demand Degradation of Bioplastics into Soil-Enriching Compost.
  2025 AIChE Annual Meeting, Boston, MA
- **Srivastava, M.**, Nader, M., Varisco, D., & Raghavan, S.R.
  Are Animal Tissues Elastic or Viscoelastic? *(Invited Poster)*
  Annual Burgers Symposium 2025, University of Maryland, MD
- **Srivastava, M.**, Nader, M., Varisco, D., & Raghavan, S.R.
  Viscoelasticity of Animal Tissues and Hydrogels. *(Poster)*
  Biomedical Engineering Society Meeting (BMES) 2024, Baltimore, MD
- **Srivastava, M.**, & Raghavan, S.R.
  How Viscoelastic are Tissues? Insights into Tissue Rheology and on Gels That Can Mimic the Same.
  Society of Rheology Meeting 2024, Austin, TX
- **Srivastava, M.**, Choudhary, H., Philip, I. M., & Raghavan, S.R.
  Can Plastics be degraded in Minutes? Yes, on-demand. *(Poster)*
  ACS Spring Meeting 2024, New Orleans, LA
- **Srivastava, M.**, & Raghavan, S.R.
  Using CO_2 to Modulate Self-Assembly and Rheology: Viscosity Change in Surfactant Fluids Induced by CO_2.
  International Congress on Rheology 2023, Athens, Greece
- **Srivastava, M.**, Ganesh, S., Subraveti, S. N., Nguyen, D., & Raghavan, S.R.
  Designing Gels to Absorb Impact—How a Thin Gel Can Protect an Egg from Breaking.
  MRS Spring Meeting 2023, San Francisco, CA
- **Srivastava, M.**, Zhou, C., & Raghavan, S.R.
  Using CO_2 to modulate self-assembly: Viscosity increase or decrease in surfactant fluids induced by CO_2.
  MRS Spring Meeting 2023, San Francisco, CA

Awards and Honors

- Kulkarni Foundation Summer Research Fellowship – *University of Maryland College Park 2025*
- “Start a SUD Startup” Challenge Winner – *National Institute on Drug Abuse (NIDA) 2023*
- Goldhaber Travel Award Recipient – *University of Maryland College Park 2023*
- Outstanding Teaching Assistant – *University of Maryland 2023*
- Vibha Gold Medal – *Indian Institute of Technology Kanpur 2019*